

# Ahmed Elmeligy, P.Eng.

Edmonton, AB

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## EDUCATION

### University of Calgary

Master of Public Policy, GPA 4.0/4.0

Calgary, AB

2023 – 2024

- **Capstone Project:** Optimizing Transmission Infrastructure for Renewable Energy Integration in Alberta.

### University of Ottawa

BASc in Chemical Engineering, Summa Cum Laude, GPA 3.97/4.0

Ottawa, ON

2013 – 2018

- **Thesis:** Artificial Neural Networks as Metamodels for the Multi-Objective Optimization of Biobutanol Production.
- Graduated with the highest academic standing in the Department of Chemical and Biological Engineering.

## PROFESSIONAL EXPERIENCE

### Government of Alberta, Executive Council – Policy Coordination Office

Manager, Business Planning and Reporting; Full-time

Edmonton, AB

Dec 2024 – Present

- **Highlight:** Oversee the development and implementation of data-driven systems to streamline mandate tracking, policy reporting, and decision-making support at the highest level of the provincial government.
- Facilitate cross-ministry collaboration by providing strategic guidance and training to enhance capacity for policy reporting and alignment with government priorities, ensuring transparent communication to senior decision-makers.

Policy Analyst; Full-time

May 2024 – Dec 2024

- **Highlight:** Enhancing policy capacity within the government by coordinating the development and delivery of public policy training materials for individual contributors and senior managers.
- Leading the data cleaning and migration of decision-making scheduling software, enhancing functionality and providing critical insights to the Premier's and Ministers' offices.
- Awarded the 2024 Premier's Public Service Award for Service Delivery Innovation as part of Policy Coordination Office, the highest level of recognition in the Alberta Public Service.

### Imperial Oil/ExxonMobil, Kearn Oil Sands

Extraction & Tailings Facilities Engineer; Fly-in/Fly-out 2-weeks on/2-weeks off

Wood Buffalo, AB

Jan 2022 – Mar 2023

- **Highlight:** Led profitability initiatives to increase unit run length by 100% thereby reducing shutdown frequency and increasing unit utilization, via design changes and metallurgy upgrades.
- Led assessments of unit operation to determine root-cause issues and barriers to profitability, safety, and reliability. Coordinated teams of operators, technologists, and maintenance personnel to identify and address areas of inefficiency.

### Imperial Oil/ExxonMobil, Strathcona Refinery

Distillation & Hydrotreating Process Engineer-in-Training; Full-time

Edmonton, AB

Aug 2020 – Jan 2022

- **Highlight:** Oversaw performance optimization efforts for process units generating a cumulative P&L of \$200M annually and achieved record asphalt production by resolving bottlenecks and longstanding reliability challenges. Provide 24/7 on call technical support, acting as a focal point and escalation contact for high priority technical issues.
- Tapped by management to lead a \$52M Crude Unit improvement project and was recognized for consistently challenging existing practices to uncover process improvements.

Process Development Engineer-in-Training; Full-time

Jan 2019 – Jul 2020

- **Highlight:** Designed and oversaw the execution of a new injection facility in response to off-spec bunker fuel, bringing regulatory compliance standards in-line with expectations to prevent a potential \$9M in annual penalties.
- Led early-stage project planning and capital budgeting of crude and fluid catalytic cracking units margin projects to determine feasibility and profitability (NPV, DCF, IRR), ultimately delivering net benefits of \$21M per year.

Process Control Engineering Co-op; Full-time

Jan 2016 – Dec 2016

- **Highlight:** Identified vessels susceptible to brittle fracture and developed the logic for smart alarm implementation onto the Distributed Control Systems (DCS), to prevent cold safety events.
- Analyzed historical process data of fired equipment and developed correlations for statistical dynamic fuel gas clamps to mitigate furnace flooding/bogging risks.

## RESEARCH EXPERIENCE

**Natural Sciences and Engineering Research Council of Canada (NSERC) REDEVELOP Program** **Calgary, AB**  
Circular Economy Team Lead and Researcher Nov 2023 – May 2024

- Researched the co-development of geological CO<sub>2</sub> sequestration with geothermal energy extraction and lithium brine recovery in Alberta.

**University of Ottawa, Centre for Research in Photonics** **Ottawa, ON**  
Research Assistant May 2018 – Aug 2018

- Designed and built a portable microfluidic system utilized for the detection of disease in biofluids using long-range surface plasmon-polariton waveguides.

**Tianjin University, Key Laboratory of Membrane Science and Desalination Technology** **Tianjin, China**  
Visiting Research Assistant May 2017 – Aug 2017

- Researched, fabricated, and characterized polymer matrices for membrane distillation using non-destructive evaluation methods: XRD, SEM, TEM, and FTIR.

**University of Ottawa, ENERGY - S.M.A.R.T. Group** **Ottawa, ON**  
Research Assistant Sept 2014 - Apr 2015

- Conducted research on adsorbents recovery ability of carbon dioxide to recycle to produce hydrocarbon fuel, which is an alternative to carbon capture and sequestration (CCS).

## LEADERSHIP EXPERIENCE

**Imperial Oil (Edmonton) United Way Campaign Chair** Jan 2021 – Nov 2021

- Provided leadership and direction to 15 committee members raising a total of \$338K, surpassing campaign goals. Awarded Employee Campaign Chair of The Year by United Way of The Alberta Capital Region.

**Imperial Oil (Edmonton) United Way Campaign Co-Chair and Events Lead** Jun 2020 – Dec 2020

- Owned the ideation, planning and execution of 7 new virtual and physical events to raise a total of \$332K, surpassing campaign goals. Nominated for Outstanding Committee Member of the Year.

**CoV-Informed (Risk and Strategy Consulting) Co-Founder** May 2020 – Oct 2021

- Provided strategy consulting to small businesses by applying data analytics to provide insights. Selected as a top team amongst 1800+ competitors from 89 countries in MIT's Beat the Pandemic Hackathon.

**Strathcona Working from Home Engagement Committee Lead** Mar 2020 – Oct 2020

- Engaged 500+ employees at Strathcona Refinery by starting employee well-being initiatives using a collaboration-centered approach to combat loneliness and disengagement as a result of the COVID-19 pandemic.

**Khair4All (Poverty Alleviation Charity) Co-Founder and Director** Aug 2011 – June 2017

- Administered a budget of \$150K and managed the daily activities of 50+ volunteers across fundraising and operations.

## COMMUNITY INVOLVEMENT & EXTRACURRICULAR ACTIVITIES

- Patient Visitor, Covenant Health Mar 2025 – Present
- Board Member - Communications Coordinator, Alpine Club of Canada Edmonton Section Feb 2025 – Present
- Mealtime Companion, Bethany Riverview Sept 2022 – Apr 2024
- Food Distribution Volunteer, MFNS Food Bank Mar 2022 – Apr 2024
- The Fraser Institute Student Ambassadorship Feb 2024 – Mar 2024
- Awards Committee Member, Graduate Student Association, University of Calgary Jan 2024 – Mar 2024
- Virtual Math & English Tutor, YYC Generosity Apr 2021 – Nov 2021
- Responder, Edmonton Regional Search and Rescue Association Jan 2020 – Nov 2021
- Patient Visitor & Pandemic Support Volunteer, Covenant Health Jul 2019 – Nov 2021
- Treasurer, TM4PM Toastmasters Club Oct 2019 – Sep 2020
- Meal Prep Volunteer, Hope Mission Sept 2018 – Sept 2019
- Student-Athlete, University of Ottawa Novice Rower Sep 2017 – Mar 2018
- President, Society of Environmental Engineering, Development, Management and Science Sep 2017 – May 2018
- Elected Engineering Faculty Council Member, University of Ottawa Sep 2015 – Apr 2016
- Research and Development Specialist, Enactus Canada Sep 2015 – May 2016

## PUBLICATIONS

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- G. Fellows, **A. Elmeligy**, K. Koch, R. Mansell, D. Rama, I. Skendrovic “Energy Transitions Roundtable Proceedings”, School of Public Policy Paper Series, August 2024.
- **A. Elmeligy**, M. Bailey “Optimizing Transmission Infrastructure for Renewable Energy Integration in Alberta”, School of Public Policy Master of Public Policy Capstone Project, August 2024.
- **A. Elmeligy**, P. Mehrani, and J. Thibault “Artificial Neural Networks as Metamodels for the Multi-Objective Optimization of Biobutanol Production”, Applied Sciences (ISSN 2076-3417), June 2018.

## COMPETENCIES

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**Programming & Automation:** VBA, Python, Java, R, C, Fortran.

**Modeling & Engineering Systems:** Aspen HYSYS/UniSim, Honeywell Control Language, Minitab, LabVIEW.

**Enterprise Tools:** Microsoft Office Suite, Microsoft Dynamics, SharePoint, Jira, Power BI.

**AI & Productivity Platforms:** Microsoft Copilot (incl. Copilot Studio agents), Google Gemini, ChatGPT.

## CERTIFICATIONS

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|--------------------------------------------------|----------------------------------------|------------------------------------------------------|
| • Private Pilot License - Aeroplane              | • Intermediate First Aid & Level C CPR | • Indigenous Relations Training Program              |
| • Night Rating                                   | • Wilderness Advanced First Aid        | • Citizens' Police Academy (Edmonton Police Service) |
| • Restricted Operator Certificate - Aeronautical | • Mental Health First Aid              | • ASME B31.3 Code Design                             |
| • Incident Command System 100                    | • Avalanche Skills Training 1          | • Slurry Multi-Species PipeFlow Model                |
| • Accelerated Freefall Program                   | • Industrial Firefighting Fundamentals |                                                      |

## AWARDS

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- 2024 Premier's Public Service Award for Service Delivery Innovation
- 2024 REDEVELOP Conference – Placed 1<sup>st</sup> out of 3 teams (\$1,000)
- 2023 National Sciences and Engineering Research Council of Canada's (NSERC) CREATE Grant (\$7,130)
- 2023 Giving Day Graduate Scholarship in the School of Public Policy (\$5,000)
- 2021 United Way Alberta Capital Region Employee Campaign Chair of The Year (Private Sector)
- 2020 United Way Alberta Capital Region Outstanding Committee Member of the Year Nominee
- 2020 MIT Beat the Pandemic II Hackathon Finalist (\$1,500)
- 2019 & 2020 Imperial Oil Strathcona Excellence Award (People and Culture)
- 2018 Canada Hatch Plant Design Competition - Placed 2<sup>nd</sup> out of 22 universities across Canada (\$1,000)
- 2018 Canadian Society of Chemical Industry Award (Engineering)
- 2018 Faculty Plaque in Chemical Engineering
- 2017 International Experience Scholarship (\$6,000)
- 2016 University of Ottawa Merit Scholarship (\$1,000)
- 2015 – 2017 Chemical Engineering Professors' Scholarship (\$250 x3)
- 2014 Undergraduate Research Opportunity Program (UROP) Award (\$1,000)
- 2013 – 2018 University of Ottawa's Renewable Admission Scholarship (\$4,000 x4)
- 2013 – 2018 Dean's Honour List
- 2013 Dean's Merit Scholarship (\$5,000)

## HOBBIES & INTERESTS

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**Adventure Pursuits:** Aviation (private pilot), ultra marathons, mountaineering, cross-country skiing, downhill skiing, backcountry hiking, rock and ice climbing, biking, skydiving, scuba diving.

**Sports:** Flag football (uOttawa intramurals), rowing (uOttawa men's roster), rugby (Saudi Arabia national rugby league team, Jeddah Rugby Football Club Juniors Under-19), football (Jeddah Jaws), soccer (high school varsity), Brazilian jiu-jitsu, boxing.

**Creative Interests:** Photography (landscape, architecture, portrait), blogging, vlogging, reading, chess.